

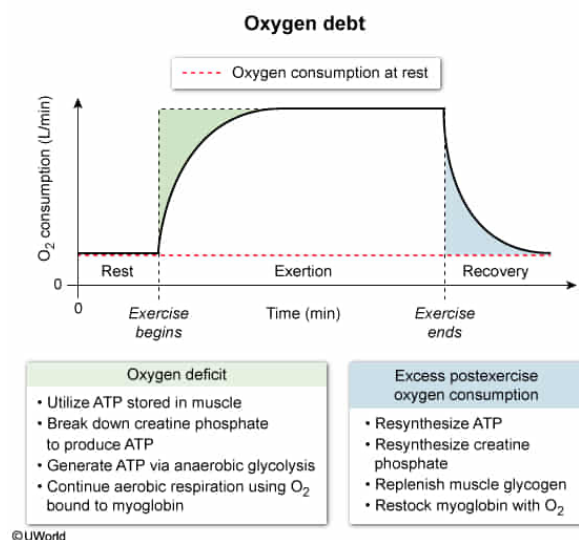
Excess post-exercise oxygen consumption is the extra amount of oxygen consumed after the completion of exercise and before respiration returns to resting levels. This extra oxygen is used to restore muscle fibers to their pre-exercise state by doing all of the following EXCEPT:

- ☐ A. replenishing the ATP stores within the muscle fiber. [7%]
- ✓ ☒ B. facilitating metabolic reactions to lower pH in the cytosol of the muscle fiber. [62%]
- ☐ C. restocking the myoglobin O₂ stores within the muscle fiber. [4%]
- ☐ D. supplying the ATP used to replenish glycogen stores in the muscle fiber. [25%]

Correct

61% answered correctly

Explanation:



Excess post-exercise oxygen consumption (EPOC), or **oxygen debt**, represents the amount of oxygen necessary to bring a muscle back to its resting state. EPOC is the difference between the amount of oxygen required for the physical activity and the total amount of oxygen used.

Muscles **require ATP** to contract. **Aerobic respiration** uses oxygen to generate more molecules of ATP per metabolized glucose molecule than **anaerobic glycolysis**. Consequently, aerobic respiration is the primary source of ATP in muscle fibers during vigorous, long-term exercise. However, when oxygen delivery from the blood is insufficient (eg, during rapid, high-intensity exercise) or the demand for contraction exceeds the speed at which aerobic respiration can generate ATP, muscle fibers compensate by:

1. utilizing ATP that is stored in the muscle fiber.

2. breaking down creatine phosphate, a high-energy molecule stored by muscle cells during periods of rest, to generate ATP.
3. generating ATP via anaerobic glycolysis, which yields 2 pyruvate molecules and 2 net ATP molecules per glucose molecule. In the absence of oxygen, the 2 pyruvate molecules are converted to lactic acid.
4. continuing aerobic respiration using oxygen that is bound to myoglobin, a red-pigmented oxygen storage molecule.

Once motor neuron input ceases, the fiber relaxes and begins replenishing these energy reserves. Consequently, oxygen intake remains elevated even after physical activity has stopped because oxygen is required to resynthesize ATP and creatine phosphate (**Choice A**) and restock myoglobin saturation with oxygen (**Choice C**).

In addition, glycogen stores **broken down** during exercise must be replenished. In a process known as the **Cori cycle**, lactic acid is transported to the liver, where ATP is used to metabolize lactic acid back to glucose. This and other glucose in the circulation is transported into muscle and stored as glycogen (**Choice D**) or metabolized to **pyruvate**.

However, the increased demand for **ATP hydrolysis** (which produces protons) during exercise (*not* rest) leads to *decreased pH* and an increased reliance on anaerobic metabolic reactions in muscle. Therefore, these types of reactions would *not* require oxygen *nor* return muscle to a *pre-exercise* state.

Educational objective:

Oxygen intake remains elevated after exercise due to the increased demand in muscle cells for oxygen to replenish ATP, creatine phosphate, and glycogen stores and restock myoglobin with oxygen. The amount of additional oxygen consumed is called the excess post-exercise oxygen consumption, or oxygen debt.

Time spent:0 sec
Subject:
Biology

QID:401523
System:
Musculoskeletal System